

22. Sharpening of Image Using Laplacian Mask with Positive Center Coefficient

Aim

To perform **image sharpening using a Laplacian mask with a positive center coefficient** and enhance the edges and fine details of the image.

Laplacian Mask

The mask shown in the question is:

The center coefficient is **+5**, so the mask directly produces the sharpened image.

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Procedure

1. Read the input image.
2. Convert the image into grayscale.
3. Define the Laplacian sharpening mask.
4. Apply the mask to the image using convolution.
5. Clip the resulting pixel values to the range **0–255**.
6. Display the original and sharpened images.
7. Save the sharpened image.

Python Code

```
import cv2

import numpy as np

import matplotlib.pyplot as plt
```

```
# Read input image

img = cv2.imread("input.jpg")

if img is None:

    print("Error: Image not found")

    exit()

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Laplacian mask with positive center coefficient

mask = np.array([

    [0, -1, 0],

    [-1, 5, -1],

    [0, -1, 0]

], dtype=np.float32)

# Apply the mask

sharpened = cv2.filter2D(gray, -1, mask)

# Clip pixel values

sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

# Display images
```

```
plt.figure(figsize=(10, 4))
```

```
plt.subplot(1, 2, 1)
```

```
plt.imshow(gray, cmap="gray")
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
```

```
plt.imshow(sharpened, cmap="gray")
```

```
plt.title("Sharpened Image")
```

```
plt.axis("off")
```

```
plt.show()
```

```
# Save output
```

```
cv2.imwrite("laplacian_positive_output.jpg", sharpened)
```

Input

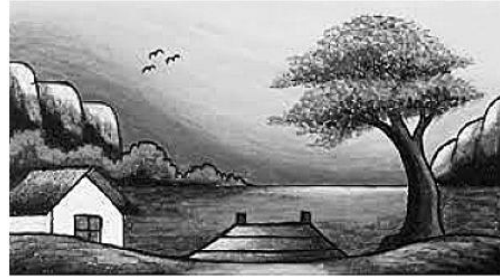


Output

Original Image



Sharpened Image



Result

Thus, the image was successfully sharpened using the **Laplacian mask with a positive center coefficient (+5)**, resulting in enhanced edges and image details.